

JEE MAIN REVISION MASTER: CAPACITOR 2026 2

QUESTIONS 3.33% WEIGHTAGE

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 17 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Capacitance & Dielectric Insertion	17 PYQs	Capacitance
Moderate	Capacitor Networks & Energy Redistribution	7 PYQs	Circuits & Energy

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Capacitance & Dielectric Insertion	$C_0 = \frac{\epsilon_0 A}{d}$, $\text{quad } C = \frac{K\epsilon_0 A}{d}$, $\text{quad } C_{\text{slab}} = \frac{\epsilon_0 A}{d - t(1 - 1/K)}$, $\text{quad } q_p = q\left(1 - \frac{1}{K}\right)$
Capacitor Networks & Energy Redistribution	$C_{\text{parallel}} = \sum C_i$, $\text{quad } \frac{1}{C_{\text{series}}} = \sum \frac{1}{C_i}$, $\text{quad } V_{\text{common}} = \frac{C_1 V_1 + C_2 V_2}{C_1 + C_2}$, $\text{quad } \Delta U_{\text{loss}} = \frac{C_1 C_2}{2(C_1 + C_2)}(V_1 - V_2)^2$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Capacitance & Dielectric Insertion (Very High Yield — 17 PYQs)

Core Principle: Parallel plate capacitance, dielectric polarization, and partial dielectric slab insertion.

■ **Caution / JEE Pitfall:** If battery remains **CONNECTED**: $V = \text{const}$, C increases by K $\implies$ Q increases by K . If battery is **DISCONNECTED**: $Q = \text{const}$ $\implies$ V decreases by $1/K$.

■ **Speed Trick:** Energy stored $U = \frac{1}{2}CV^2 = \frac{Q^2}{2C}$. When dielectric is inserted with disconnected battery, U decreases because electrostatic attraction does work.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "In the circuit shown in the figure, the potential difference across the $4\mu\text{F}$ capacitor is..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "A parallel plate capacitor has a uniform electric field 'E' in the space between the plates. If the distance between the plates is 'd...' [Ans: C]
- [2024 Shift] Q#3 (MCQ): "A capacitor of capacitance $5\mu\text{F}$ is charged by a battery of emf 10 V . At an instant of time, the potential difference across the capacit..." [Ans: N/A]

3.2 Capacitor Networks & Energy Redistribution (Moderate Yield — 7 PYQs)

Core Principle: Equivalent capacitance reduction, charge sharing between interconnected capacitors, and Joule heating loss.

■ **Caution / JEE Pitfall:** When connecting with opposite polarity, use numerator $(C_1 V_1 - C_2 V_2)$ for common potential.

■ **Speed Trick:** Energy loss during charge sharing $\Delta U_{\text{(loss)}}$ is completely independent of wire resistance; it always turns into heat and EM radiation.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "In the circuit shown in the figure, the potential difference across the $4\mu\text{F}$ capacitor is..." **[Ans: N/A]**
- **[2023 Shift] Q#4 (MCQ):** "Five capacitance each of value $1\mu\text{F}$ are connected as shown in the figure. The equivalent capacitance between A and B is..." **[Ans: N/A]**
- **[2020 Shift] Q#3 (MCQ):** "The difference between equivalent capacitances of two identical capacitors connected in parallel to that in series is $6\mu\text{F}$. The value o..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

In the circuit shown in the figure, the potential difference across the $4\mu\text{F}$ capacitor is

- (A) 3 V
- (B) 4 V
- (C) 9 V
- (D) 12 V

Correct Answer: None

Solution Analysis: The $9\mu\text{F}$ and $3\mu\text{F}$ capacitors are connected in parallel. In parallel combination, capacitances are added directly. $C_{(p)} = 9 + 3 = 12\mu\text{F}$ Now this $12\mu\text{F}$ equivalent capacitor is in series with the $4\mu\text{F}$ capacitor. For two capacitors in series, the equivalent capacitance is: $C_{(eq)} = (4 \times 12 / 4 + 12) = (48 / 16) = 3\mu\text{F}$ The battery voltage is 12V. So the total charge supplied by the battery is: $Q = C_{(eq)} \times V = 3\mu\text{F} \times 12\text{V} = 36\mu\text{C}$ In a series combination, each capacitor has the same charge. So the charge